



Aalto University

On nonstationary subspace analysis for multivariate random fields

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Agenda of the Presentation

Motivation and model definition

Population version of the estimator

Sample version of the estimator

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Linear blind source separation for random fields

Let $\mathbf{u} \in \mathbb{R}^p$, and let x be a d -dimensional random field on $\mathbb{R}^p$. Consider a model

$$\begin{pmatrix} x_1(\mathbf{u}) \\ \vdots \\ x_d(\mathbf{u}) \end{pmatrix} = x(\mathbf{u}) = \mathbf{A}z(\mathbf{u}) = \mathbf{A} \begin{pmatrix} z_1(\mathbf{u}) \\ \vdots \\ z_d(\mathbf{u}) \end{pmatrix},$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is invertible and does not depend on $\mathbf{u}$.

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$\Rightarrow$ We observe x but z has simpler dependence structure. The goal is to extract $\mathbf{A}$ or z .

- Components of $z(\mathbf{u})$ are uncorrelated and *stationary* (Bachoc, Genton, et al. 2020).
- Components of $z(\mathbf{u})$ are independent and *nonstationary* (Bachoc, Muehlmann, et al. 2024).

Our model

Let $\mathbf{u} \in \mathbb{R}^p$, and let x be a d -dimensional random field on $\mathcal{U} \subset \mathbb{R}^p$. Consider the model

$$x(\mathbf{u}) = \mathbf{A}z(\mathbf{u}) = \begin{pmatrix} \mathbf{A}_\eta & \mathbf{A}_s \end{pmatrix} \begin{pmatrix} \eta(\mathbf{u}) \\ s(\mathbf{u}) \end{pmatrix},$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is invertible and does not depend on $\mathbf{p}$.

■ $\mathbf{A}_\eta \in \mathbb{R}^{d \times q}$ and $\mathbf{A}_s \in \mathbb{R}^{d \times (d-q)}$.

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- $\mathbf{A}_\eta \in \mathbb{R}^{d \times q}$ and $\mathbf{A}_s \in \mathbb{R}^{d \times (d-q)}$.
- Components of q -dimensional field $\eta(\mathbf{u})$ are *nonstationary*.
- Components of $(d - q)$ -dimensional field $s(\mathbf{u})$ are *stationary*.
- The random fields η and s are independent.

For simplicity of the presentation, we only consider nonstationarity in the mean.

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Local and global statistics (1)

Divide the domain $\mathcal{U} \subset \mathbb{R}^p$ into blocks $\mathcal{S}_j$. Sample in block $\mathcal{S}_j$ is S_j .

- Number of blocks is k . Index blocks by $j \in \{1, \dots, k\}$.
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- We denote that S_0 is the total sample and n_0 is the total sample size.
- Idea for estimation
 - $\Rightarrow$ Compare local statistics to global statistics.

Local and global statistics (2)

Define the following local (and global) statistics

$$\mathbf{m}_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} \mathbb{E} (x(\mathbf{u})) ,$$

$$\text{Cov}_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} \mathbb{E} ((x(\mathbf{u}) - m_j(x)) (x(\mathbf{u}) - m_j(x))^{\top})$$

For identifiability assume...

$$\mathbb{E}(s(\mathbf{u})) = \mathbf{0}, \text{Cov}(s(\mathbf{u})) = \mathbf{I}, m_0(\eta) = \mathbf{0} \text{ and } \text{Cov}_0(\eta) = \mathbf{I}.$$

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“Location” and “covariance” fixed: It follows that $m_0(z) = \mathbf{0}$ and $\text{Cov}_0(z) = \mathbf{I}$.

Scatter based on blocks

Define

$$M_{k,n}(x) = \frac{1}{k} \sum_{j=1}^k (\mathbf{m}_j(x) - \mathbf{m}_0(x)) (\mathbf{m}_j(x) - \mathbf{m}_0(x))^{\top}$$

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Idea:

- For stationary s we have $M_{k,n}(s) = \mathbf{0}$.
- For nonstationary η we have $M_{k,n}(\eta) \neq \mathbf{0}$ is symmetric positive definite.

Extracting the unmixing matrix

1. Whiten the field: $\tilde{\mathbf{x}}(\mathbf{u}, t) = \mathbf{W}(\mathbf{x}(\mathbf{u}, t) - m_0(\mathbf{x}))$, where $\mathbf{W}\text{Cov}_0(\mathbf{x})\mathbf{W}^\top = \mathbf{I}$.

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2. Diagonalize: Let $\mathbf{V}_{k,n} \in \mathbb{R}^{d \times d}$ be a matrix such that

$$\mathbf{V}_{k,n}\mathbf{V}_{k,n}^\top = \mathbf{I} \quad \text{and} \quad \mathbf{V}_{k,n}^\top \mathbf{M}_{k,n}(\tilde{\mathbf{x}})\mathbf{V}_{k,n} = \mathbf{\Lambda}_{k,n},$$

where $\mathbf{\Lambda}_{k,n} \in \mathbb{R}^{d \times d}$ is a diagonal matrix with diagonal elements in nonincreasing order.

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where $\mathbf{\Lambda}_{k,n} \in \mathbb{R}^{d \times d}$ is a diagonal matrix with diagonal elements in nonincreasing order.

3. Transform back: an unmixing matrix $\mathbf{\Omega}_{k,n} \in \mathbb{R}^{d \times d}$ based on $M_{k,n}$ is defined as $\mathbf{\Omega}_{k,n} = \mathbf{V}_{k,n}^\top \mathbf{W}$.

Identifiability: definition

The unmixing problem based on $M_{k,n}$ is identifiable if for every matrix $\Omega_{k,n}$ satisfying definition of the previous slide there exist invertible matrices $\mathbf{B}_\eta \in \mathbb{R}^{q \times q}$ and $\mathbf{B}_s \in \mathbb{R}^{(d-q) \times (d-q)}$ such that

$$\Omega_{k,n} = \begin{pmatrix} \Omega_\eta \\ \Omega_s \end{pmatrix} = \begin{pmatrix} \mathbf{B}_\eta & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_s \end{pmatrix} \mathbf{A}^{-1}, \quad \Omega_\eta \in \mathbb{R}^{q \times d}, \quad \Omega_s \in \mathbb{R}^{(d-q) \times d}.$$

Identifiability: meaning

We obtain “mixed” stationary and nonstationary signals:

- $\Omega_{\eta}x(\mathbf{u}) = \mathbf{B}_{\eta}\eta(\mathbf{u})$

- $\Omega_s x(\mathbf{u}) = \mathbf{B}_s s(\mathbf{u})$

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We obtain “mixed” stationary and nonstationary signals:

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- $\Omega_s \mathbf{x}(\mathbf{u}) = \mathbf{B}_s \mathbf{s}(\mathbf{u})$

We can choose $\mathbf{B}_\eta$ and $\mathbf{B}_s$ to be orthogonal:

- $\text{Cov}(\mathbf{B}_s \mathbf{s}(\mathbf{u})) = \mathbf{I}$

- For example, under $\text{Cov}(\eta(\mathbf{u})) = \sigma^2(\mathbf{u})\mathbf{I}$, we have that $\text{Cov}(\mathbf{B}_\eta \eta(\mathbf{u})) = \sigma^2(\mathbf{u})\mathbf{I}$

Identifiability: condition

- Let $\lambda_{\eta}^{(\min)}$ be the smallest eigenvalue of $M_{k,n}(\eta)$
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Meaning: all the components of η are nonstationary in the mean is sufficient.

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Above we focused on scatter based on $m_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} \mathbb{E}(x(\mathbf{u}))$.

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Define,

$$\hat{m}_j(x) = \frac{1}{n} \sum_{\mathbf{u} \in S_j} x(\mathbf{u}, t),$$

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and

$$\hat{M}_{k,n}(x) = \frac{1}{k} \sum_{j=1}^k (\hat{m}_j(x) - \hat{m}_0(x)) (\hat{m}_j(x) - \hat{m}_0(x))^{\top}$$

Our asymptotic framework

- Sample size per block and number of blocks: $n \rightarrow \infty$ and $k \rightarrow \infty$ ($n_0 = nk \rightarrow \infty$).

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- Sample size per block and number of blocks: $n \rightarrow \infty$ and $k \rightarrow \infty$ ($n_0 = nk \rightarrow \infty$).
- Increasing domain: $\text{Leb}(\mathcal{U}) = \text{Leb}(\mathcal{U}_{n_0}) \rightarrow \infty$, as $n_0 \rightarrow \infty$.
- Suppose that the smallest eigenvalue $\lambda_{\eta,k,n}^{(min)}$ of $M_{k,n}(\eta)$ satisfies

$$\liminf_{k,n \rightarrow \infty} \lambda_{\eta,k,n}^{(min)} \geq c, \quad c > 0.$$

Asymptotic identifiability

If we have

$$\left\| \hat{M}_{k,n}(z^{st}) - M_{k,n}(z) \right\| = O_{\mathbb{P}}(r_{k,n}),$$

and that

$$\left\| \widehat{\text{Cov}}_0(\mathbf{z}) - \mathbf{I} \right\| = O_{\mathbb{P}}\left(\frac{1}{\sqrt{n_0}}\right),$$

where $1/\sqrt{n_0} = O(r_{k,n})$, then there exists a sequence of random invertible matrices $\hat{\mathbf{B}}_{\eta}^{(k,n)} \in \mathbb{R}^{q \times q}$ and $\hat{\mathbf{B}}_s^{(k,n)} \in \mathbb{R}^{(d-q) \times (d-q)}$ such that

$$\left\| \begin{pmatrix} \hat{\mathbf{B}}_{\eta}^{(k,n)} & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{B}}_s^{(k,n)} \end{pmatrix} \hat{\boldsymbol{\Omega}}_{k,n}(\mathbf{x}) - \mathbf{A}^{-1} \right\| = O_{\mathbb{P}}(r_{k,n}).$$

Other assumptions

- **Finite dependence:** Let $U_1, U_2 \subset \mathcal{U}$. There exists a fixed $L > 0$ not depending on the sets U_1 and U_2 such that if $\min \{ \| \mathbf{u} - \mathbf{u}' \| : \mathbf{u} \in U_1, \mathbf{u}' \in U_2 \} > L$ then the sets $\{z(\mathbf{u}) : \mathbf{u} \in U_1\}$ and $\{z(\mathbf{u}) : \mathbf{u} \in U_2\}$ are independent.

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- **Uniformly bounded fourth moments:** It holds that $\sup_{i \in [n_0]} \max_{j \in [d]} \mathbb{E} |z_j(\mathbf{u}_i)|^4 \leq C < \infty$ uniformly for all n_0 .

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- **Increasing domain:** There exists a fixed $\Delta > 0$ such that for all $i, j \in [n_0]$, $i \neq j$, we have $\|\mathbf{u}_i - \mathbf{u}_j\| \geq \Delta$.

The main result

We have

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- For $\sqrt{n_0}$ -consistent estimation you need $k/n = O(1)$.
- The frameworks
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can fail.

Link to slides: [https://github.com/perej1/presentations/
tree/main/nonstationary-field](https://github.com/perej1/presentations/tree/main/nonstationary-field)

References

- Bachoc, François, Marc G Genton, et al. (2020). “Spatial blind source separation”. In: *Biometrika* 107.3, pp. 627–646. DOI: 10.1093/biomet/asz079.
- Bachoc, François, Christoph Muehlmann, et al. (2024). “Large-sample properties of non-stationary source separation for Gaussian signals”. In: *Electronic journal of statistics* 18.1.